

ENGINEERING SPECIFICATIONS

Document Name: VFD Set Points- KWI With Danfoss make VFD(FC102)

Document No:	ESP-15-16-110	Revision: R11 (05/01/2023)	(UPDATED FOR LOGO & MODBUS PARITY)	Date: 15/12/2020
Prepared By: RDT	Revised By: IGP	Approved By: SKP		Sheet: 1 of 2
ID	Motor Parameters	Value		
		Selection A	Selection B	Selection C
0 30	Custom Readout Unit	RPM	RPM	RPM
0 31	Custom Readout Min Value	0	0	0
0 32	Custom Readout Max Value	3600	3600	3600
0 60	Main Menu password	2020	2020	2020
0 61	Access to main menu	LCP read only	LCP read only	LCP read only
0 40	[Hand on] Key on LCP	Disabled	Disabled	Disabled
1 03	Torque Characteristics	Compressor torque	Compressor torque	Compressor torque
1 20	Motor power (KW)	106	106	160
1 22	Motor Voltage	400	400	400
1 23	Motor Frequency	60	60	60
1 24	Motor Current	173	173	265
1 25	Motor Nominal Speed	3509	3509	3549
3 03	Maximum Reference	53	60	60
3 13	Reference Site	Remote	Remote	Remote
3 15	Reference 1 Source	Analog input 53	Analog input 53	Analog input 53
3 16	Reference 2 Source	No function	No function	No function
3 17	Reference 3 Source	No function	No function	No function
3 41	Ramp 1 Ramp Up Time	8	8	8
3 42	Ramp 1 Ramp Down Time	10	10	10
4 10	Motor Speed Direction	Clockwise	Clockwise	Clockwise
4 12	Motor Speed Low Limit [Hz]	22	22	22
4 14	Motor Speed High Limit [Hz]	53	60	60
4 16	Torque Limit Motor Mode	120	120	120
4 18	Current Limit	$\left[\frac{\text{RLA of Comp (as per SO)}}{\text{Motor Current (para. 1 24)}} \times 100 \right] + 10$	$\left[\frac{\text{RLA of Comp (as per SO)}}{\text{Motor Current (para. 1 24)}} \times 100 \right] + 10$	$\left[\frac{\text{RLA of Comp (as per SO)}}{\text{Motor Current (para. 1 24)}} \times 100 \right] + 10$
4 19	Max Output Frequency	54	61	61
4 50	Warning Current Low	0	0	0
4 51	Warning Current High	Showing Auto	Showing Auto	Showing Auto
4 52	Warning Speed Low	0	0	0
4 53	Warning Speed High	3509	3509	3549
5 13	Terminal 29 Digital Input	No operation	No operation	No operation
5 40	Function Relay	Drive ready	Drive ready	Drive ready
5 40 1	Function Relay	Running	Running	Running
6 15	Terminal 53 High Ref./ Feedback Value	53	60	60
6 50	Terminal 42 Output	Motor cur. 0-Imax	Motor cur. 0-Imax	Motor cur. 0-Imax
8 01	Control Site	Digital and ctrl.word	Digital and ctrl.word	Digital and ctrl.word
8 02	Control Source	FC Port	FC Port	FC Port
8 10	Control Profile	FC profile	FC profile	FC profile
8 30	Protocol	Modbus RTU	Modbus RTU	Modbus RTU
8 31	Address	1,2 (As per compressor/Circuit number)	1,2 (As per compressor/Circuit number)	1,2 (As per compressor/Circuit number)
8 32	Baud Rate	19200 Baud	19200 Baud	19200 Baud
8 33	Parity / Stop Bits	No parity , 1 Stop Bit (carel controller)	No parity , 1 Stop Bit (carel controller)	No parity , 1 Stop Bit (carel controller)
		Even parity, 1 Stop Bit (Siemens controller)	Even parity, 1 Stop Bit (Siemens controller)	Even parity, 1 Stop Bit (Siemens controller)
8 43 0	PCD Read Configuration	Status Word [1603]	Status Word [1603]	Status Word [1603]
8 43 1	PCD Read Configuration	Main Actual Value [%] [1605]	Main Actual Value [%] [1605]	Main Actual Value [%] [1605]
8 43 2	PCD Read Configuration	Reference [%] [1602]	Reference [%] [1602]	Reference [%] [1602]
8 43 3	PCD Read Configuration	Power [kW] [1610]	Power [kW] [1610]	Power [kW] [1610]
8 43 4	PCD Read Configuration	Motor voltage [1612]	Motor voltage [1612]	Motor voltage [1612]
8 43 5	PCD Read Configuration	Frequency [1613]	Frequency [1613]	Frequency [1613]
8 43 6	PCD Read Configuration	Motor current [1614]	Motor current [1614]	Motor current [1614]
8 43 7	PCD Read Configuration	DC Link Voltage [1630]	DC Link Voltage [1630]	DC Link Voltage [1630]
8 43 8	PCD Read Configuration	Heatsink Temp. [1634]	Heatsink Temp. [1634]	Heatsink Temp. [1634]
14 25	Trip Delay at Torque Limit	25	25	25
14 26	Trip Delay at Inverter Fault	2	2	2
16 37	Inverter max current	Showing Auto	Showing Auto	Showing Auto
22 43	Wake-up Speed [Hz]	30	30	30

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NOTE				
**	FLA SETPOINT IN CONTROLLER			
	1) FOR SELECTION 'A' - RLA(AS PER SO/ CHILER NAMEPLATE) X 0.76			
	2) FOR SELECTION 'B' & 'C' - RLA(AS PER SO/ CHILER NAMEPLATE) X 0.87			
SR NO	MODEL	CIRCUIT I	CIRCUIT II	
1	KWI130.14	A	NA	
2	KWI135.14	A	NA	
3	KWI140.14	A	NA	
4	KWI145.14	B	NA	
5	KWI150.14	B	NA	
6	KWI155.14	B	NA	
7	KWI170.14/KWI130.1G	C	NA	
8	KWI175.14/KWI145.1G	C	NA	
9	KWI180.14/KWI170.1G	C	NA	
10	KWI195.14/KWI195.1G	C	NA	
11	KWI200.14/KWI210.1G	C	NA	
12	KWI205.14	C	NA	
13	KWI230.14	C	NA	
14	KWI235.14	C	NA	
15	KWI240.14	C	NA	
16	KWI248.14	C	NA	
17	KWI250.14	C	NA	
18	KWI255.14	C	NA	
19	KWI260.14	C	NA	
20	KWI268.14	C	NA	
21	KWI270.14	C	NA	
22	KWI280.14	C	NA	
23	KWI285.14	C	NA	
24	KWI290.14	C	NA	
25	KWI265.24	A	A	
26	KWI270.24	A	A	
27	KWI275.24	A	A	
28	KWI298.24	B	B	
29	KWI300.24	B	B	
30	KWI310.24	B	B	
31	KWI320.24	B	C	
32	KWI325.24	B	C	
33	KWI335.24	B	C	
34	KWI345.24/KWI260.2G	C	C	
35	KWI350.24/KWI275.2G	C	C	
36	KWI355.24/KWI295.2G	C	C	
37	KWI365.24/KWI320.2G	C	C	
38	KWI375.24/KWI340.2G	C	C	
39	KWI380.24/KWI355.2G	C	C	
40	KWI395.24/KWI370.2G	C	C	
41	KWI400.24/KWI380.2G	C	C	
42	KWI405.24/KWI390.2G	C	C	
43	KWI425.24/KWI405.2G	C	C	
44	KWI430.24/KWI420.2G	C	C	
45	KWI435.24	C	C	
46	KWI460.24	C	C	
47	KWI465.24	C	C	
48	KWI470.24	C	C	
49	KWI480.24	C	C	
50	KWI485.24	C	C	
51	KWI490.24	C	C	
52	KWI500.24	C	C	
53	KWI510.24	C	C	
54	KWI515.24	C	C	
55	KWI518.24	C	C	
56	KWI520.24	C	C	
57	KWI525.24	C	C	
58	KWI530.24	C	C	
59	KWI535.24	C	C	
60	KWI540.24	C	C	
61	KWI550.24	C	C	
62	KWI555.24	C	C	
63	KWI560.24	C	C	
64	KWI565.24	C	C	
65	KWI575.24	C	C	
66	KWI580.24	C	C	